STUDENT MANAGEMENT SYSTEM

A Console-Based Application Using C Programming

Project Title	Student Management System
Programming Language	C (C99)
Submitted by	_____
Class / Semester	_____
College	_____
Submitted to	_____
Date	_____

1. Introduction

The Student Management System is a menu-driven console application developed using the C programming language. It helps a user maintain basic student information such as student ID, name, age, course and marks. The program uses file handling to save records locally so that information is available again when the application is reopened.

2. Problem Statement

Maintaining student details manually can take time and may lead to misplaced or inconsistent records. This project demonstrates how a simple computer program can organize student records and provide basic operations to add, display, search, modify and remove information.

3. Objectives

• To develop a simple student-record application using C.

• To practise structures, functions, loops, conditional statements and file handling.

• To store and retrieve records from a local file.

• To provide a clear menu for common record-management operations.

• To validate basic user input and prevent duplicate student IDs.

4. Software and Hardware Requirements

Type	Requirement
Software	C99-compatible compiler such as GCC, Clang or MinGW
Libraries	Standard C library only; no external libraries
Operating system	Windows, Linux or macOS
Hardware	Computer capable of running a C compiler

5. Main Modules

Module	Description
Add Student	Accepts student details and saves a new record. Student IDs must be unique.
View Students	Displays all saved student records in a formatted table.
Search Student	Finds and displays a record using its student ID.
Update Student	Rewrites the record file with the selected student's updated details.
Delete Student	Removes the selected record after confirmation.

6. Concepts Used

The project uses a structure to group student fields, functions to divide the program into manageable modules, a switch statement to process menu choices, loops to repeat the menu and read records, conditional statements to validate choices, and file handling functions such as fopen, fread, fwrite, fclose, remove and rename.

7. How to Compile and Run

Save the source file as **student_management.c**. Open a terminal in the folder containing the file and compile it with:

```
gcc -std=c99 -Wall -Wextra -pedantic student_management.c -o student_management
```

On Linux or macOS, run `./student_management`. On Windows with MinGW, run `student_management.exe`. The program creates **students.dat** automatically when the first record is saved.

8. Sample Test Cases

Test	Action	Expected result
1	Add a student with a new ID	Record is saved successfully.
2	Add another student with the same ID	Duplicate ID message is displayed.
3	View records	All saved records are displayed.
4	Search for an existing ID	Matching student details are displayed.
5	Search for a missing ID	Not-found message is displayed.
6	Enter marks outside 0–100	Program asks for valid marks.

9. Limitations

This is a beginner-level educational project. It runs locally and does not include a graphical interface, multiple user accounts, password protection, database integration or network access. Records are stored in a binary file that should be backed up and used with the same program structure.

10. Future Enhancements

Possible improvements include a graphical user interface, sorting and filtering, attendance management, grade calculation, CSV export, login roles and integration with a database such as SQLite or MySQL.

11. Conclusion

The Student Management System demonstrates how C programming concepts can be combined to solve a basic record-management problem. By completing this project, a learner gains practical experience with structures, functions, input validation, control statements and file handling.

Declaration: Update the cover-page details and review the program with your instructor before submission. Run the program and capture your own output screenshots if your college requires them.